NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-27/320B
19 January 1989
SUPERSEDING
MIL-T-27/320A
14 January 1983

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 160 VOLT-AMPERES, 60 HERTZ, TF4S03YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.

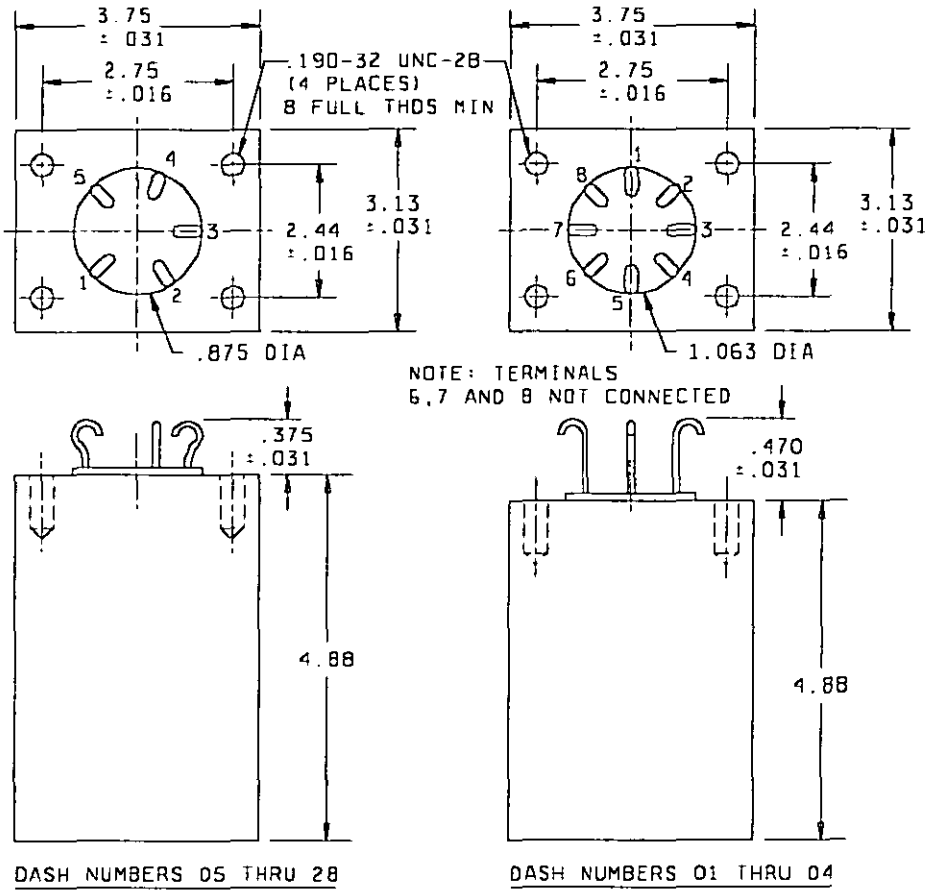


FIGURE 1. Dimensions and configuration.

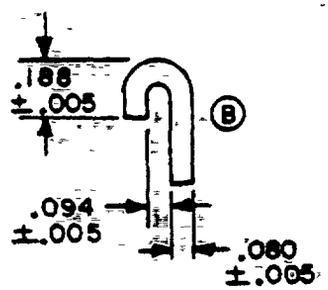
(B) denotes change

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

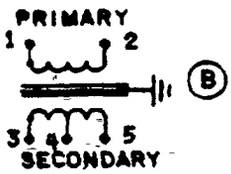
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DASH NUMBERS 05 THRU 28



DASH NUMBERS 01 THRU 04



CIRCUIT DIAGRAM

Inches	mm	Inches	mm
.005	0.13	.375	9.52
.016	0.41	.470	11.94
.031	0.79	.875	22.22
.062	1.57	1.063	27.00
.070	1.78	2.44	62.0
.080	2.03	2.75	69.8
.094	2.39	3.13	79.5
.188	4.78	3.75	95.2
.190	4.83	4.88	123.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified, tolerance is ±.063 (1.60 mm).
5. Electrical values shall be marked as specified in table I.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS. (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts ac, 50 to 60 hertz.

Primary dc resistance: 1.6 ohms ±25 percent.

Secondary voltage (3-5): See table I.

Volt-ampere rating: 160 volt-amperes.

Working voltage (1-2): 535 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Terminals: Solderable terminals.

Dash numbers	Terminal type	Height (inches)	Diameter (inches)
01 through 04	hook	0.470 ±.031	0.080 ±.005
05 through 28	hook	0.375 ±.031	0.062 ±.005

Weight: 8.2 pounds, maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 10,000 feet.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding) (at sea level): 1,500 volts rms.

Electrical characteristics:

Rated load: With 115 volts ac and 60 Hz across (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

Shall not exceed 10 percent.

Polarity: Additive with terminals 2 and 3 connected.

Temperature rise: +35°C with 115 volts ac, 60 hertz across (1-2) at an ambient temperature of +95°C, full load terminals (3-5).

Marking location: See figure 1.

Part or identifying number (PIN): M27/320-(dash number from table I).

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TABLE I. Electrical ratings.

Dash number	Secondary (3-5)		
	Voltage (volts) ±3 percent	Current 1/ (amperes)	DC resistance (ohms) ±25 percent
01	8.0	20.00	.014
02	10.0	16.00	.018
03	12.6	12.70	.038
04	14.0	11.40	.042
05	16.0	10.00	.046
06	18.0	8.89	.060
07	20.0	8.00	.070
08	22.0	7.27	.092
09	25.0	6.40	0.110
10	28.0	5.72	0.150
11	31.0	5.17	0.166
12	35.0	4.57	0.240
13	39.0	4.10	0.270
14	43.0	3.70	0.382
15	48.0	3.33	0.405
16	55.0	2.91	0.600
17	63.0	2.52	0.640
18	70.0	2.280	0.93
19	80.0	2.000	1.06
20	90.0	1.780	1.49
21	100	1.600	1.69
22	110	1.450	2.30
23	123	1.300	2.57
24	138	1.150	3.60
25	157	1.020	4.10
26	175	0.914	5.82
27	200	0.800	6.70
28	220	0.728	8.95

1/ For 50 Hz operation, reduce secondary current by 10 percent.

QUALITY ASSURANCE PROVISIONS:

Extent of qualification:

Qualification testing and approval to M27/312-26, M27/316-30, and M27/322-24 shall be sufficient to grant qualification approval to MIL-T-27/312 through MIL-T-27/322, inclusive, all parts.

Qualification test and approval to M27/320-28 shall be sufficient to grant qualification approval M27/320-01 through M27/320-28.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0725-03)

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